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Proposal for:
ICF
9300 Lee Highway
Fairfax, VA 22031

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Proposal Number:
QU-0039

Introduction

Thank you for the opportunity to provide a proposal to determine the technical and economic feasibility of a district ground source heat pump (GSHP) system for the Missoula MRL project in Missoula, MT.

Understanding of project

The City of Missoula is preparing to develop approximately 12.6 acres of land. The proposed project is to include a mixed-use development of approximately 400,000 square feet of multi-family residential buildings, townhomes, and some commercial / retail space.

The intent is to use electricity to provide heating, cooling, and domestic hot water (DHW) for the development rather than rely on fossil fuels for heating and DHW, and electricity for cooling. The city wishes to explore the potential of creating a district-scale ground heat exchanger (GHX) for the project that can be owned and operated by the city or by a third party that is prepared to own and operate the system as a geothermal utility.

Proposed design methodology

Energy modeling

The size, cost, and performance of a GHX is sensitive to the peak heat transfer to and from the ground. It is also sensitive to the total heat rejection to the ground when the building is being cooled compared to the amount of heat extracted from the ground to heat the building which is defined as the annual energy balance.

Accurate hourly energy modeling of the building(s) connected to a ground heat exchanger (GHX) is critical to the development of a GSHP system that is cost-effective to build, performs efficiently and predictably for the life of the project and is easily maintained. Energy loads are determined by simulating building performance based on the architectural and mechanical system design, anticipated occupancy, and the historical hourly weather data for the location.

Working with the architectural and mechanical design team in the early stages of the design process provide the opportunity to adjust energy loads to and from the GHX. Hourly energy modeling estimates the impact of

changes to glass specifications, lighting and lighting controls, ventilation air strategy, domestic hot water production, distribution system design, etc. Changes to the building energy loads can result in a reduction in the size and cost of the GHX and an improvement in the efficiency and performance of the system. GEOptimize will typically develop several iterations of the hourly energy model for a building to determine the impact on the size, cost and performance of the GHX.

Energy efficiency measures resulting in improvement in GHX performance and a reduction in capital cost are reviewed with the design team and owners. Preliminary GHX modeling based on approved changes in the energy model is developed based on site constraints and measured geological conditions to ensure the required land area is available and it can be built cost-effectively with the contractor resources available in the area.

GEOptimize uses Trane Trace 700 to simulate the energy performance of buildings and to create hourly energy model load profiles that will be imported into GHX design software for further simulation.

GHX modeling

The performance of a GHX is dependent on:

- Total annual heat transfer to and from the building(s)
- Peak instantaneous heat transfer to and from the building(s)
- Ambient earth temperature at the location
- Thermal characteristics of the earth in which the GHX is constructed
- Detailed design and configuration of the GHX

Thermal properties of the soil can be estimated with reasonable accuracy if information about the geological conditions is available. Information about the geological conditions is often available on water well databases at State operated websites, or in interviews with drilling contractors and / or geologists familiar with the location.

The land area available for construction of a GHX is determined with feedback from the client and development design team. The size and configuration of the land area available for construction of a GHX may determine the depth of vertical boreholes required to meet the heating and cooling loads of the project, or if another type of GHX may be appropriate on the site.

GEOptimize uses Ground Loop Design (GLD) software to simulate the performance of vertical and horizontal GHX's.

District system modeling

Different types of buildings have distinctly different hourly energy load profiles. A school, for example, is typically occupied during a weekday, unoccupied on the weekends and evenings and almost always unoccupied in summer. A residential building is occupied heavily from late afternoon till the following morning, and less heavily occupied during a weekday, while more heavily occupied on a weekend. Peak heating and cooling loads in the buildings do not coincide.

Connecting buildings with different energy load profiles to a system that can transfer energy from a building being cooled to another building requiring heat.

GHX's absorb energy rejected from a building when it is being cooled. Soil and rock are poor heat conductors. Heat absorbed by the earth surrounding the GHX piping takes time to dissipate. The energy is stored in the soil around the pipe until it dissipates or is drawn from the ground when heat is needed in the building. The ground stores excess heat when it's not required in other buildings and releases it when it's not available from other buildings.

TRNSYS is a software that can be used to simulate the performance of thermal and electrical energy systems. Simulating the performance of the district GSHP system with TRNSYS can be used to optimize GHX module sizing, energy transfer pipe sizing and flow rates required to move energy through the buildings in the development.

Information required

The following information is required to accurately model the heating and cooling loads of the home and to predict the performance of the GHX:

- Estimate of the number, size and types of buildings that are proposed for this development
- Preliminary layout of the proposed development
- Information about the type of system that would be installed in the buildings in a "business as usual" scenario
- Gas and electricity detailed rate structure including time of use rates and demand penalties (if applicable)

Scope of work

To accurately assess the feasibility of a district geothermal system in the MRL development, GEOOptimize will follow the design methodology summarized above. In practical terms, this will include:

- Information gathering video calls with representatives from the City of Missoula, ICF and the developer / design team selected to create the proposed development. This will be required to determine the type, size, layout of the site, and construction of buildings proposed for the development. This will be used to create representative buildings for each of the buildings proposed for the development. Additional discussions may be required with individual groups to ensure adequate information if available for development of accurate hourly energy models.
- Iterative hourly energy models will be developed to balance energy load profiles. This may include, but will not be limited to, energy models with a range of energy efficiency measures.
- An aggregated hourly energy load profile will be developed using the hourly energy load profiles of individual buildings.
- A review of the State of Montana water well data base, interviews with well drilling and geothermal well drilling contractors and geologists familiar with the area in and surrounding Missoula will be conducted to estimate the thermal properties and most cost-effective drilling depth at the site.
- Aggregated hourly energy loads and estimated geological conditions will be used to determine the size of GHX required to meet the heating and cooling needs of the proposed development.
- A preliminary system schematic will be developed and a simulation using TRNSYS modeling software will be created to refine the preliminary system schematic.
- A video call presentation with project stakeholders to present the preliminary findings to estimate the viability of a district geothermal system in the community. Feedback will be solicited from the project stakeholders.
- An economic analysis based on the revised / updated conceptual design will include:

- Estimated energy cost of GSHP district system in comparison with “business as usual” scenario.
 - Estimated budget estimates of the incremental cost of constructing proposed district geothermal system.
 - Estimated maintenance concerns of operating a district geothermal system
- Environmental benefits of the district geothermal system in comparison to a “business as usual” scenario, including CO₂ emissions based on the carbon intensity of electricity in Montana. If renewable energy is available for the project (either on site generation or purchased renewable energy credits), this will be incorporated into the calculations.
- Feedback from project stakeholders will be incorporated into the simulation software to ensure the proposed system reflects the requirements of the City of Missoula, the design team and developers.
- Revisions based on feedback from project stakeholders will be incorporated into an updated presentation and a written report will be prepared summarizing the results of the study.

Pricing

Description	Rate (USD)	Qty (hrs)	Line Total
Feasibility assessment for MRL Project, Missoula, MT (as described in Scope of Work)			
Estimated hours at (Engineer 1):	\$175	80	\$14,000.00
Estimated hours at (Engineer 2):	\$200	150	\$30,000.00
Estimated hours at (Analyst 5):	\$250	60	\$15,000.00
Total fee proposal:		290	\$59,000.00

Terms

Additional work out of scope will be charged at \$200/hr. Invoices will be submitted monthly for work completed to date. Payment is expected on receipt of invoice.

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